PathForge — Session Summary (2026-03-12)

Overview

Today's work integrated the **pre-computed ML sample data** into the PathForge platform and completed the **dynamic career state** system. The platform went from using 5 universities with fake alignment scores to **22 real East African universities**, **55 programs**, and **1,430 ML-backed alignment records**.

What Changed

1. `PathForgeBackend/prisma/seed.js` — Complete Rewrite (424 lines)

Before: 5 universities, 6 programs, simple string-matching `computeAlignment()` formula, ~50 manually curated skills.

After: Reads directly from the ML-generated CSV files in `prisma/Sample data/`:

What	Before	After
Universities	5	**22** (Kenya, Uganda, Tanzania, Rwanda)
Programs	6	**55** (14 program types)
Alignment records	~156 (formula)	**1,430** (pre-computed ML scores)
Curriculum per program	Hand-crafted	Derived from `program_vectors.csv` (top 10 skills)
Alignment scoring	String match	Sigmoid-normalized ML scores from `alignment_matrix.csv`

Key additions:

- CSV parsing helpers (`readCSV` , `splitLine`) to load sample data at seed time
- Skill name formatter (`fmtSkill`) mapping 50+ snake_case skill IDs → display names via `master_skill_dictionary.csv`
- Sigmoid normalization (`normScore`) converting raw ML scores (0–0.25 range) to user-friendly 0–1 range
- Proxy scoring for Software Engineer & Data Engineer (not in original ML model) — averages scores from similar careers
- Batch insertion of alignment records (batch size: 300)

---

2. `Pathforge-Frontend/lib/api.ts` — Complete Rewrite (331 lines)

Before: Static MOCK data with fake similar careers ("Business Intelligence Analyst", "Research Scientist", "Quantitative Analyst") and hardcoded Data Scientist gap analysis.

After: Full 24×24 career similarity matrix embedded from `career_similarity.csv` , plus dynamic mock generators.

Key additions:

- `_SIM_NAMES` — 24 career names array
- `_SIM` — 24×24 cosine similarity matrix (576 values)
- `_SIM_PROXIES` — Maps Software Engineer → Full Stack Developer, Data Engineer → Data Scientist for similarity lookups
- `getMockSimilarCareers(career, topN)` — Looks up any career in the similarity matrix, returns top-N most similar with sector info
- `getMockGapResult(career)` — Generates a dynamic gap analysis from the career's required skills (no more hardcoded Data Scientist)
- `MOCK.universityMatches` updated to real universities (Nelson Mandela AIST, CMU Rwanda, Makerere, ALU)
- `MOCK.similarCareers` updated to real Data Scientist similarities (ML Engineer 0.877, AI Engineer 0.714, etc.)

- `SkillGap` type imported and properly typed (fixed TypeScript errors)

---

3. `Pathforge-Frontend/app/universities/page.tsx` — Updated (390 lines)

Changes:

- `UNIVERSITY_WEBSITES` expanded from **5 → 22** universities with verified URLs:
  - Kenya: UoN, JKUAT, Kenyatta, Moi, TU Kenya, USIU Africa
  - Uganda: Makerere, Kyambogo, UCU, UMU, MUST
  - Tanzania: UDSM, Ardhi, Mzumbe, NM-AIST, MUHAS, UDOM
  - Rwanda: UR, CMU Rwanda, ALU, UoK, AUCA
- `PROGRAMS` filter expanded from 4 → **14 program types** (CS, IT, SE, DS, AI, Cybersecurity, BBA, Accounting, Finance, Economics, Analytics, EE, Environmental Eng, Industrial Eng)
- `TABLE_DATA` updated from old universities to sample data universities (CMU Rwanda, Makerere, UoN)

---

4. `Pathforge-Frontend/app/dashboard/page.tsx` — Updated (366 lines)

Changes:

- Added imports for `getMockSimilarCareers` and `getMockGapResult`
- **Similar careers fallback** now uses `getMockSimilarCareers(targetCareer, 3)` instead of static MOCK — shows real similar careers for any selected career
- **Gap analysis fallback** now uses `getMockGapResult(targetCareer)` — generates career-specific skill gaps instead of always showing Data Scientist gaps

---

5. `Pathforge-Frontend/lib/careerStore.ts` — Created (24 lines)

(Created in earlier session today)

Simple localStorage wrapper for persisting career state across pages:

- `getTargetCareer()` / `setTargetCareer(career)` — stores selected career
- `getStudentProfile()` / `setStudentProfile(profile)` — stores assessment results
- SSR-safe with `typeof window` checks
- Default career: "Data Scientist"

---

6. `Pathforge-Frontend/lib/types.ts` — Updated (75 lines)

(Updated in earlier session today)

Added/refined TypeScript interfaces:

- `SkillGap` — with severity union type `"critical" | "moderate" | "minor" | "none"`
- `GapAnalysisResult`, `SimilarCareer`, `StudentProfile`, `MarketData`, `LearningModule`, `ProgressPoint`, `UserState`

---

7. Multiple Frontend Pages — Updated (earlier session)

All pages updated to read career from `careerStore` instead of hardcoding "Data Scientist":

Page	File	Change
Assessment	`app/assessment/page.tsx`	Saves career + profile to localStorage
Dashboard	`app/dashboard/page.tsx`	Reads career from store
Skill Gap	`app/skill-gap/page.tsx`	Reads career + profile from store

Progress	`app/progress/page.tsx`	Dynamic career display
Profile	`app/profile/page.tsx`	Dropdown with all 26 careers
Market Intel	`app/market-intel/page.tsx`	Dynamic career initialization
Roadmap	`app/roadmap/page.tsx`	Dynamic learning modules
Universities	`app/universities/page.tsx`	Dynamic career for alignment query
Layout	`app/layout.tsx`	`suppressHydrationWarning` on body

Bug Fixes

Issue	Fix
All pages compared against "Data Scientist" regardless of user's career	Replaced with `getTargetCareer()` from localStorage across 9 pages
Auditor assessment showed irrelevant skills (Python, TensorFlow)	Career-specific required skills now drive the assessment
MOCK similar careers were fake (didn't exist in our data)	Replaced with real 24×24 cosine similarity matrix
Hydration error from Grammarly browser extension	Added `suppressHydrationWarning` to `<body>` tag
TypeScript errors: `careerData.skills` possibly undefined	Added null coalescing `?? []` fallback
TypeScript errors: severity type mismatch	Explicit union type annotation on severity variable

---

Source Data Files Used

All from `PathForgeBackend/prisma/Sample data/` (ML model v1.0.0 output):

File	Content	Records
`program_metadata.csv`	University names, programs, regions	55 programs, 22 universities
`alignment_matrix.csv`	Career × Program alignment scores	24 × 55 = 1,320 scores
`career_similarity.csv`	Career × Career cosine similarity	24 × 24 = 576 scores
`program_vectors.csv`	156-dimensional skill vectors per program	55 programs
`master_skill_dictionary.csv`	Normalized skill names with IDs	350 skills
`career_vectors.csv`	156-dimensional skill vectors per career	24 careers
`career_metadata.csv`	Career names, IDs, sectors	24 careers

Verification Checklist

- [x] TypeScript compiles cleanly (`npx tsc --noEmit` — 0 errors)
- [ ] Database seeding (`npx prisma db push && node prisma/seed.js`) — requires PostgreSQL running
- [ ] API endpoints return ML-backed data (`/api/alignment/:career` , `/api/similarity/:career`)
- [ ] Dashboard shows real similar careers for any selected career
- [ ] Universities page displays all 22 universities with working website links

- [ ] Assessment → Dashboard → Skill Gap flow works end-to-end

---

Architecture After Changes

```
User selects career (Assessment)
        │
        ▼
  localStorage (careerStore.ts)
        │
        ├──► Dashboard ──► getCareerAlignment() ──► Backend (seed.js data)
        │                  getSimilarCareers()       └── alignment_matrix.csv
        │                  getSkillGap()             └── career_similarity.csv
        │                       │
        │                  Falls back to:
        │                  getMockSimilarCareers() ◄── 24×24 similarity matrix
        │                  getMockGapResult()      ◄── career required skills
        │
        ├──► Universities ──► getCareerAlignment() ──► 22 universities, 55 programs
        │
        ├──► Skill Gap ──► getSkillGap() with student profile
        │
        └──► All other pages read career dynamically
```

PathForge — Session Summary (2026-03-13)

Overview

This session added **five major feature areas** to the PathForge frontend: a Learning Resources browser, a fully dynamic University Comparison tool, a Market Intelligence suite with career comparison, a Skill Gap career comparison view, and a polish pass on the Student Progress Journey page.

New Features

8. `Pathforge-Frontend/lib/resources.ts` — Created (~350 lines)

Comprehensive course database covering every skill across all 26 careers (~47 unique skills).

What	Detail
Platforms covered	Coursera, edX, Udemy, YouTube, Khan Academy
Courses per skill	3–5 curated entries with real external URLs
Types	`Platform`, `Course`, `SkillResources`
Exports	`SKILL_RESOURCES`, `PLATFORM_CONFIG`, `LEVEL_CONFIG`

---

9. `Pathforge-Frontend/app/resources/page.tsx` — Created (~400 lines)

New page at `/resources?skill=<name>` reachable from the "View Resources" button on the Roadmap.

Component	Purpose

`SkillBrowser`	Grid of all 47 skills with course counts — shown when no `?skill=` param
`SkillDetail`	Left sidebar (filters: free-only toggle, level, platform) + 2-col `CourseCard` grid
`CourseCard`	Platform badge, free tag, level pill, duration, external link (`target="_blank"`)
`ResourcesContent`	Reads `?skill=Excel` from URL, routes to browser or detail view

Roadmap link updated:

```
// app/roadmap/page.tsx — before
href="/assessment"
// after
href={`/resources?skill=${encodeURIComponent(mod.title)}`}
```

---

10. `Pathforge-Frontend/lib/universities.ts` — Created (~500 lines)

Full data layer for the University Program Comparison feature.

What	Detail
Universities	15 East African institutions with full metrics
Metrics	overall, alignment, employability, roi, research, tuition, facilities, studentLife, academics, internationalRank
Helpers	`filterUniversities()`, `getMetricValue()`, `getMetricScore()`, `buildCSV()`
Radar data	Per-university radar chart values
Badges	Ranked badges ("Best ROI", "Top Research", etc.) per university

---

11. `Pathforge-Frontend/app/universities/page.tsx` — Full Rewrite (~700 lines)

Fully dynamic head-to-head University Comparison page.

Feature	Implementation
Career switcher	Updates all data dynamically
University selector modal	Grid of all 15, tick 2–4 to compare
URL-based state	`?compare=cmu,makerere&region=Rwanda` — synced on every change via `router.replace()`
Share	`navigator.clipboard.writeText(window.location.href)` + toast notification
Export CSV	`buildCSV(selected)` → Blob → download link
Comparison table	× remove buttons, ★ star highlights for best value per metric row
Radar chart	SVG radar for selected universities with `RADAR_COLORS`
All universities list	Ranked list with `+ Compare` / `✓ Added` toggles

---

12. `Pathforge-Frontend/lib/marketData.ts` — Created (~600 lines)

Market intelligence data for all 26 careers sourced from Andela Africa Report 2024, LinkedIn East Africa, ICPAK, World Bank.

What	Detail
Salary data	`salaryKES`, `salaryTrend`, `cagr`, `jobOpenings` per career
Trajectory	Entry → Mid → Senior → Lead salary stages
Regional salaries	Kenya, Rwanda, Uganda, Tanzania, Ethiopia via PPP multipliers
Skills in demand	Per-career list with trend arrows (up/stable/down), aligned to career requirements
Top employers	12 employers per sector drawn from regional employer pools
Currencies	9 currencies (KES base) with `convertSalary()` and `formatSalaryShort()` helpers

---

13. `Pathforge-Frontend/app/market-intel/page.tsx` — Full Rewrite (~1,100 lines)

Fully dynamic Market Intelligence page with single-career overview and multi-career comparison mode.

Overview Mode

Feature	Detail
Career readiness banner	Reads `getStudentProfile()` from localStorage, computes readiness vs. required skill levels, shows personalized "You're X% ready" message
KPI cards	Median salary (currency-converted), market demand, CAGR, job openings
Currency picker	9-currency dropdown — updates all salary figures live
Regional salary bar chart	SVG bar chart for Kenya / Rwanda / Uganda / Tanzania / Ethiopia
Skills in demand	Career-specific skills with trend arrows and progress bars
Trajectory chart	SVG area chart with average line + user-adjusted projection (blue dashed)
Top employers	Filterable by region
Add to Compare	Button switches to Compare tab with current career pre-loaded

Compare Mode (new)

Feature	Detail
Career picker	Sector filter tabs + color-coded selected pills (up to 4)
`COMPARE_COLORS`	`["#f97415","#3b82f6","#10b981","#8b5cf6"]`
Metrics table	12 metrics: salary, demand, CAGR, openings, trajectory stages, top skill, top employer — with ★ best-value highlights
`MultiTrajectoryChart`	SVG multi-line chart for up to 4 careers with end-point labels
Grouped bar chart	Toggle between Salary / Demand / CAGR metrics with animated bars
Regional salary table	Country-by-country comparison with ★ best markers

---

14. `Pathforge-Frontend/app/skill-gap/page.tsx` — Major Addition

Added a "Compare Careers" tab alongside the existing "My Analysis" view.

Feature	Detail
Mode toggle	`My Analysis | Compare Careers` tab switcher in header
`computeCareerReadiness()`	Computes readiness % for all 26 careers from student's localStorage profile
All careers ranked list	Sorted by user readiness %, with search, sector filter, sort dropdown
Gap badges	Critical / moderate / mastered counts inline per career
`+ Compare` button	Select up to 4 careers for focused head-to-head
Head-to-head table	Readiness %, critical/moderate/minor/mastered counts, time to ready — ★ star on best value
SVG bar chart	Readiness bars for all selected careers with `COMPARE_COLORS`
Top skills to bridge	3 most critical gaps per career shown inline
Shortcut button	"Compare to Other Careers" in the readiness donut card

---

15. `Pathforge-Frontend/app/progress/page.tsx` — Polish Pass

Complete visual and functional overhaul of the Student Progress Journey page.

What	Before	After
Readiness chart	Bar chart (invisible — CSS height bug)	Smooth SVG area + line chart with bezier curves, gradient fill
Projected trend	None	3-month dashed projection line extending from current data
Projection banner	None	"At current pace, you'll reach 90% by Est. Jul 2026" tile with linear extrapolation
Monthly velocity	Hardcoded `+2.6%`	Computed from `(latest - earliest) / months` = `+3%/mo`
Improvement %	Hardcoded `+21%`	Derived from actual PROGRESS_HISTORY values
Streak days	Hardcoded `14`	Derived as `Math.round(improvement × 0.67)`
Milestone dates	Empty string `""`	Past milestones: real month labels; upcoming: `Est. Apr 2026`, `Est. May 2026`
Career switcher	None	Dropdown with all 26 careers — re-fetches skill gap data live
"Most Improved" badge	None	Green pill on fastest-improving skill in Skill Progress list
Per-skill delta	None	`+Δ% this period` annotation under each skill bar
Export	None	"Export" button triggering `window.print()` with `@media print` stylesheet
Heatmap	Predetermined hardcoded colours	Intensity mapped to actual `streakDays` count
Donut annotation	None	`+21% since Jan` chip below the donut

Architecture After Session 2

```
User selects career (Assessment / Progress career switcher)
        │
        ▼
  localStorage (careerStore.ts)
        │
        ├──► Dashboard ──► getMockSimilarCareers() / getMockGapResult()
        │
        ├──► Skill Gap ──► My Analysis + Compare Careers (all 26 careers ranked)
        │
        ├──► Market Intel ──► Overview (personalized) + Compare (up to 4 careers)
        │                      Currency picker (9 currencies)
        │                      Regional salary (5 countries)
        │
        ├──► Universities ──► Head-to-head comparison (up to 4)
        │                      Share (URL) + Export (CSV)
        │
        ├──► Roadmap ──► View Resources ──► /resources?skill=<name>
        │                                   47 skills, 3–5 courses each
        │
        └──► Progress ──► Area chart + projection + career switcher + export
```

Files Changed — Session 2 Summary

File	Type	Lines
`lib/resources.ts`	Created	~350
`app/resources/page.tsx`	Created	~400
`app/roadmap/page.tsx`	Modified	1 line
`lib/universities.ts`	Created	~500
`app/universities/page.tsx`	Full rewrite	~700
`lib/marketData.ts`	Created	~600
`app/market-intel/page.tsx`	Full rewrite	~1,100
`app/skill-gap/page.tsx`	Major addition	~600
`app/progress/page.tsx`	Polish rewrite	~380

Total new/changed lines this session: ~5,100

Verification Checklist — Session 2

- [x] TypeScript compiles cleanly after every change (`npx tsc --noEmit` — 0 errors)
- [ ] `/resources` — skill browser shows all 47 skills; clicking a skill shows 3–5 courses
- [ ] `/resources?skill=Python` — direct URL works with filters
- [ ] `/universities` — selector picks 2–4 universities; comparison table renders; Share copies URL; Export downloads CSV
- [ ] `/market-intel` — career picker loads correct skills/salary/employers; currency switcher updates all figures

- [ ] `/market-intel` Compare tab — pick 4 careers; metrics table shows ★ highlights; multi-line chart renders
- [ ] `/skill-gap` Compare tab — all 26 careers ranked; select 4; head-to-head table appears
- [ ] `/progress` — area chart visible (not blank); career switcher changes data; Export opens print dialog